

Chapter 38

Modifying Clips in the Timeline

Once you've edited a variety of clips into the Timeline, you'll start working with them as you refine your edit. In this chapter, you'll learn simple methods of modifying clips, including resizing, splitting, shuffling, disabling, copying and pasting, and duplicating.

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Keyboard Shortcuts in This Chapter

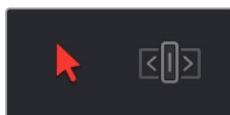
Here's a list of of keyboard shortcuts you might find helpful that relate to topics found in this chapter.

Key Shortcut	Function
A	Selection tool/mode
V	Select nearest edit to playhead
Shift-V	Select clip intersecting playhead
U	Select incoming, outgoing, or centered part of edit
Option-U	Toggle selection among video+audio, video only, and audio only
Up, Down Arrow Keys	Move selection to previous/next edit or clip
Comma, Period	“Nudge” keys to move a selected edit or clip left or right a frame at a time
Shift-Comma, Period	“Fast Nudge” keys to move a selected edit or clip left or right 5 frames at a time (customizable)
Shift-Left, Right Bracket ([,])	Trim Start to Playhead and Trim End to Playhead to trim a clip at the position of the playhead
E	Extend edit to move a selected edit point to the playhead
B	Razor blade tool for adding cuts to clips using the pointer
Command-Backslash (\)	Insert edit; adds a cut to the clip(s) at the position of the playhead
Delete	Delete clip and leave gap (lift edit)
Forward Delete	Ripple delete; deletes a clip and moves the rest of the Timeline left to fill the gap
N	Toggle timeline snapping off and on
Command-Shift-L	Toggle linked selection off and on
Command-D	Change clip duration
Command-Shift (modifier)	While dragging, holding Command-Shift lets you “shuffle” or “swap” edit clips forward or backward to rearrange clips in the Timeline
D	Disable/Enable selected clips
Command-X	Cut selected clips, leave gap

Key Shortcut	Function
Command-Shift-X	Ripple cut selection; close gap left by cut clip(s)
Command-C	Copy selected clips
Command-V	Paste clips
Command-Shift-V	Paste insert clips

Moving, Resizing, and Rolling Clips in Selection Mode

After editing a series of clips into a timeline, the next thing even the most careful of editors probably needs to do is to start making changes. The simplest changes are made in Selection Mode, using the regular arrow pointer.



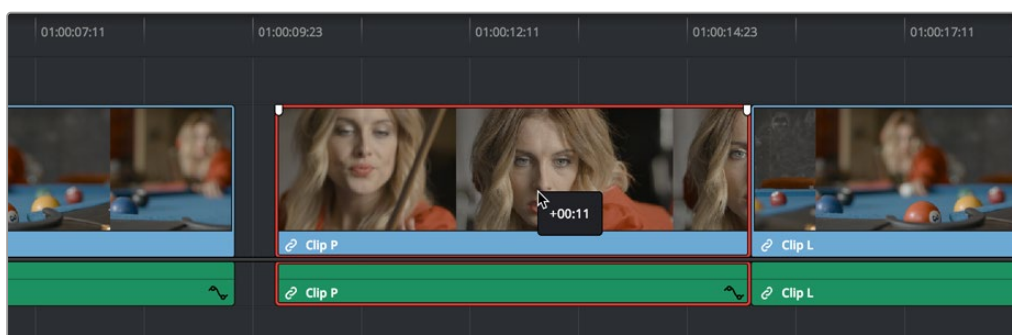
The Selection Mode button at left is enabled; the Trim mode button at right is disabled

This is the default mode when you open DaVinci Resolve, and allows you to move clips to other places in the Timeline, resize them to make them longer or shorter, and roll the edit points between two clips to move the edit to an earlier or later position on the Timeline. What this tool does depends entirely on what you click to select as you work.

Manipulating clips using the mouse:

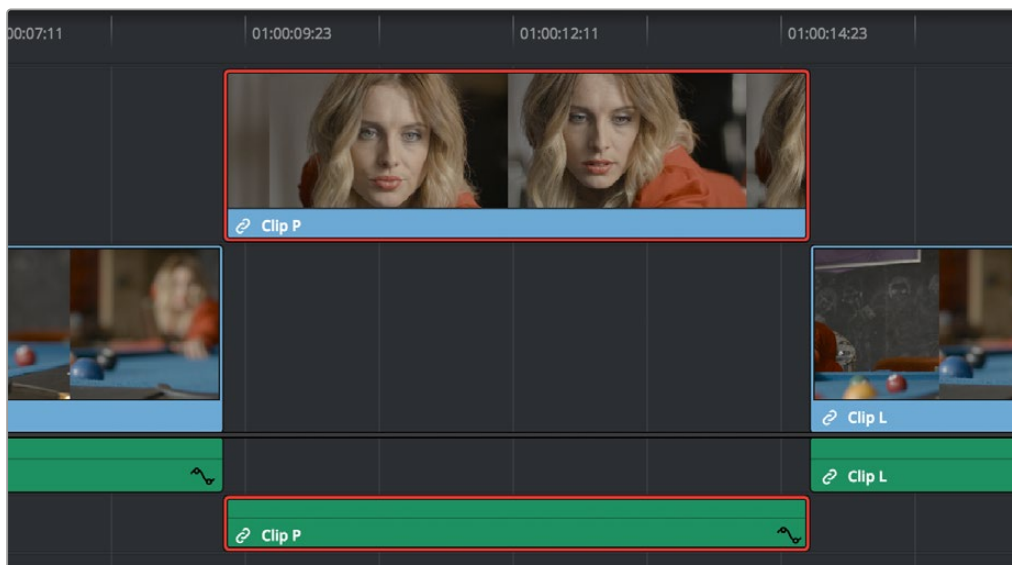
- 1 Click the Selection Mode tool (the arrow), or press A.
- 2 Do one of the following:

To move clips in the Timeline: Drag any clip in the Timeline to any other position. If you drag a clip to overlap another clip, the clip you're dragging overwrites the clip you're dropping it onto.



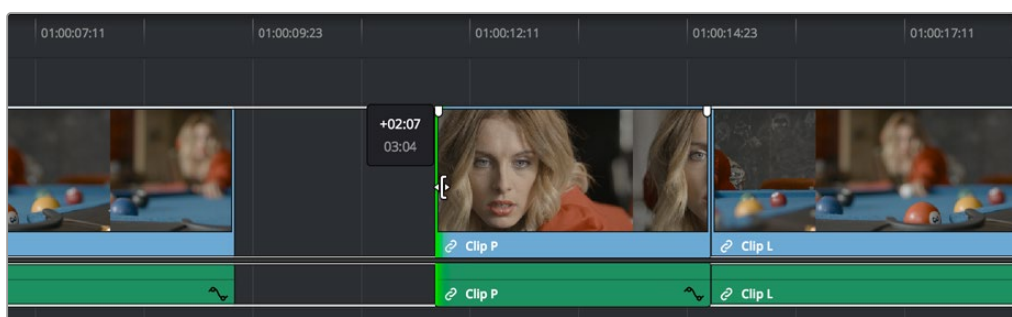
Moving a clip in the Timeline to overwrite part of another clip; a tooltip shows you how many frames you've moved

To move clips in the Timeline up or down to other tracks while keeping them at the same time: Hold the Shift key down while dragging clips up or down in the Timeline. Or, you can hold the Option key down and press Up or Down Arrow.



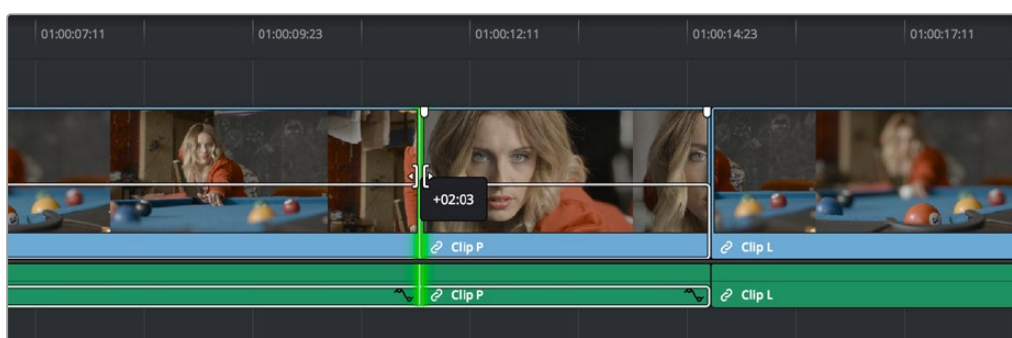
Moving a clip into another track without sliding it in time by holding the Shift key down

To shorten or lengthen clips: Move the Selection Mode pointer over the beginning or end of a clip, and when it turns into the Resize cursor, drag the In or Out point to the left or right to change the clip's length. As you do so, the audio will scrub along with the Resize cursor.



Resizing a clip in the Timeline to create a gap; a tooltip shows the offset, and outlines show you how much media is available in the clip being adjusted

To roll any edit: Move the Selection Mode pointer over any edit point, and when it turns into the Roll Edit cursor, drag it to the left or right to move the edit point while simultaneously resizing the outgoing and incoming edits points of the two clips surrounding it. As you do so, the audio will scrub along with the right clip's In point.



Rolling an edit; a tooltip shows the offset, and an outline shows the available area you can roll within

Manipulating clips using the keyboard:

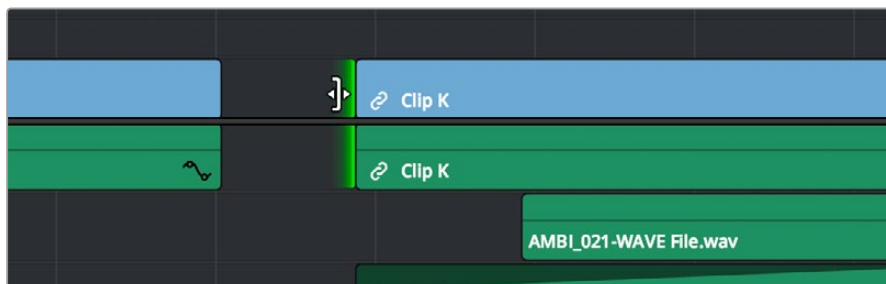
- 1 Press A to choose Selection Mode.
- 2 Do one of the following:
 - **To roll any edit incrementally:** Select the closest edit point to the playhead using the V key, moving the selection to another edit, if necessary, by using the Up Arrow and Down Arrow keys. Then press the Comma key (nudge 1 frame left) or Period key (nudge 1 frame right) to roll the selected edit to the left or right. Shift-Comma and Shift-Period nudges by 5 frames.
 - **To roll any edit using the playhead:** Select the closest edit point to the playhead using the V key, moving the selection to another edit, if necessary, by using the Up Arrow and Down Arrow keys. Then use the JKL keys to move the playhead to the frame you want to move the edit to, and press E to do an “extend” edit.
 - **To shorten or lengthen clips incrementally:** Select the closest edit point to the playhead using the V key, then use the U key to toggle the selection among the end of the outgoing clip and the beginning of the incoming clip. Then, press the Comma key (nudge 1 frame left) or Period key (nudge 1 frame right) to shorten or lengthen that side of the clip. If you nudge one end of a clip to overlap another, the clip you’re nudging overwrites the adjacent clip. Shift-Comma and Shift-Period nudges by 5 frames. In Selection mode, this either leaves a gap or overwrites neighboring clips.
 - **To shorten clips using the playhead:** Use the JKL keys to move the playhead over the frame in the Timeline where you want to set a new In or Out point for that clip, then press Shift-Left Bracket ([) to “trim start,” or Shift-Right Bracket (]) to “trim end.” No selection is necessary. In Selection mode, this leaves a gap.
 - **To lengthen clips using the playhead:** Select the closest edit point to the playhead using the V key, then use the U key to toggle the selection among the end of the outgoing clip and the beginning of the incoming clip. Then, use the JKL keys to move the playhead to the frame to want to extend that edit point to, and press E to do an “extend” edit. In Selection mode, this overwrites neighboring clips.
 - **To move clips forward or back in the Timeline:** To select a clip in preparation for moving it, either click it, or use the Spacebar or JKL keys to move the playhead over it and press Shift-V. Then press the Comma key (nudge 1 frame left) or Period key (nudge 1 frame right) to move the clip to the left or right. If you nudge a clip to overlap another clip, the clip you’re nudging overwrites the adjacent clip. Shift-Comma and Shift-Period nudges by 5 frames. In Selection mode, this leaves a gap.
 - **To move clips up or down to other tracks:** To select a clip in preparation for moving it, either click it, or use the Spacebar or JKL keys to move the playhead over it and press Shift-V. Then, press Option-Up Arrow to move the Video and Audio of that clip to the next higher-numbered track, or press Option-Down Arrow to move the Video and Audio to the next lower-numbered track.

TIP: You can hold down the Shift key while nudging a selection to do a “fast nudge.” The duration of a fast nudge is customizable in the Editing panel of the User Preferences. By default it’s five frames, but you can set it to whatever you want.

Trimming Gaps

The start and end of gaps can also be rippled using the Trim tool.

For more information, see *Chapter 44, “Trimming.”*



Using the Trim tool to ripple the Out point of a gap to narrow it

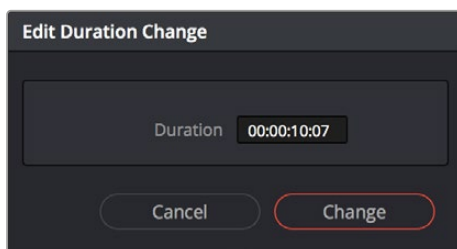
Modifying Clip Duration Via Timecode

You can change a clip's duration numerically in one of two ways.

To change a selected clip's duration:

- 1 Decide if you want to ripple the Timeline or overwrite neighboring clips when you change a clip's duration. If you want to ripple the Timeline, choose the Trim tool. If you want to overwrite neighboring clips or leave a gap, choose the Selection tool.
- 2 Do one of the following:
 - Select a clip and choose Clip > Change Clip Duration.
 - Right-click any clip in the Timeline and choose Change Clip Duration from the contextual menu.
- 3 When the Edit Duration Change dialog appears, enter a new duration in the Timecode field, and click Change.

For more information on timecode entry, refer to “Moving the Playhead Using Timecode” see *Chapter 35, “Preparing Clips for Editing and Viewer Playback.”*



A window for changing the duration of a clip in the Timeline

Resizing or Trimming Clips in the Source Viewer

You can also open a clip from the Timeline into the Source Viewer to perform trimming in different ways. You can do this in one of two ways:

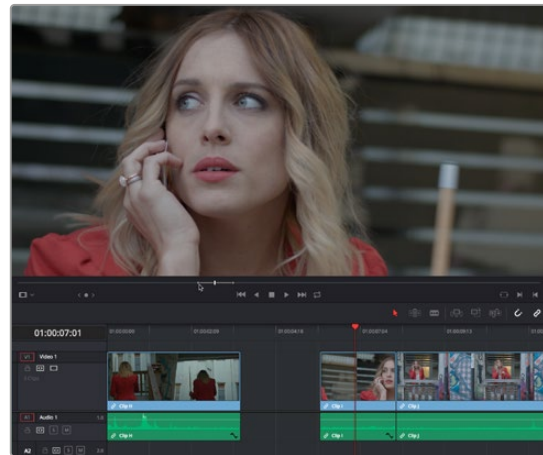
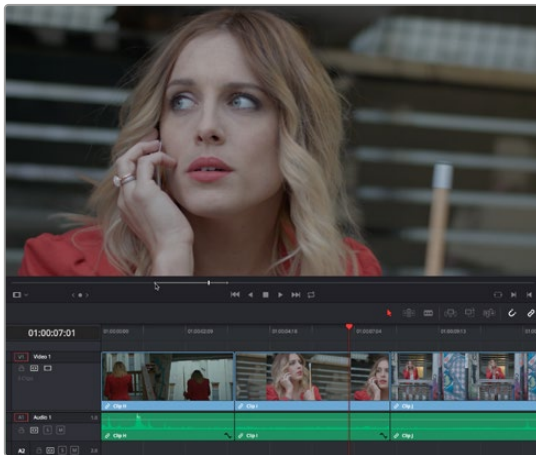
- Double-click a clip in the Timeline to open it into the Source Viewer.
- Move the playhead over a clip in the Timeline, press Shift-V to select that clip, and then press the Return or Enter key to open it into the Source Viewer.

Once you open a clip into the Source Viewer, the Source Viewer has focus, enabling you to use the Spacebar or JKL commands to move the playhead around in the Viewer in order to make edits. How these edits affect the Timeline depends on whether you use the Selection tool or the Trim tool.

TIP: To instead open a match frame of a clip in the Timeline using the pointer, hold the Option key down while double-clicking a clip.

Using the Selection Tool

When the Selection tool is selected, you can drag the In and Out markers, or use the playhead and I and O keyboard shortcuts to resize that clip in the Timeline.



(Left) A Timeline clip opened in the Source Viewer, (Right) resizing the clip and leaving a gap by dragging its In point in the Source Viewer

Using the Trim Tool

When the Trim tool is selected, dragging the In and Out points, or setting new ones using the I and O keys resizes the clip while rippling the Timeline left or right as necessary.

Doing a Slip Edit in the Viewer

If you hold the Shift key down while dragging the In or Out point of a timeline clip you've opened in the Source Viewer, you'll move both the In and Out points together, doing a slip edit of that clip's content in the Timeline. This works using either the Selection or Edit tools.

Shuffle/Swap Insert Edits

A Shuffle Insert edit (sometimes referred to as a Swap Insert edit) lets you quickly rearrange one or more selected clips in the Timeline simply by Command-Shift dragging them to the left or right. When you do so, the surrounding clips automatically move to the right or left to switch places with the clip or clips that you're dragging. This is a really fast way to reorder clips to try out different arrangements, without needing to drag clips onto multiple tracks to get them out of the way, first.

You have a lot of flexibility in how you shuffle clips around. You can select one clip, or multiple consecutive clips to shuffle. If you select multiple consecutive clips, they'll move together as a single block. You can even select multiple consecutive clips on multiple tracks to shuffle around the Timeline as a single item.

Furthermore, you can also select clips that are part of split edits, where the audio and video In and Out points start or end at different frames. In this case, how other clips move in the Timeline to make room for the split edit clip you're dragging depends on whether you click the video or audio portion of the clip to start dragging:

- If you click-and-drag the video portion of the clip, then all clips will rearrange themselves based on the duration of that video item on that track, so that all video items on that track rearrange themselves without either overwriting one another or leaving gaps. As you drag to shuffle the selection through the Timeline, overlapping linked audio items will either overwrite the audio on neighboring clips, or leave a gap.
- If you click-and-drag the audio portion of the clip, then all clips will rearrange themselves based on the duration of that audio item on that track, so that all audio items on that track rearrange themselves without either overwriting one another or leaving gaps. As you drag to shuffle the selection through the Timeline, overlapping linked video items will either overwrite the video on neighboring clips, or leave a gap.

Given the rules previously described, shuffling clips is really easy, and you can do so in one of two ways.

To shuffle insert clips with adjacent clips in the Timeline:

- 1 Turn snapping on.
- 2 Select one or more consecutive clips you want to shuffle.
- 3 Press and hold the Command and Shift keys down and drag either the video or audio portion of the selected clips to the left or right.

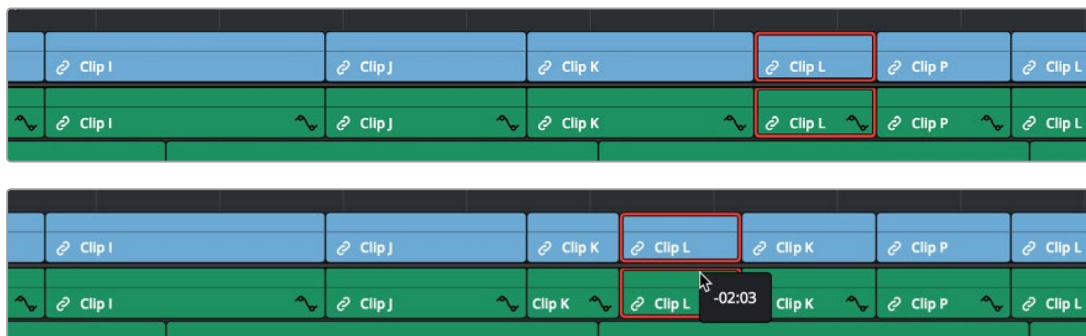
So long as you move clips to the In or Out points of adjacent clips, they'll automatically switch places with the selection of clips you're dragging. Snapping will help make sure that you align clip(s) you're dragging with previously existing edit points until dropped in the desired location.



Before and after clip L being shuffled with clips I, K, and J in a scene to rearrange the sequence

To shuffle insert one clip into adjacent clips in the Timeline:

- 1 Turn snapping off.
- 2 Select one or more consecutive clips in the Timeline that you want to shuffle.
- 3 Press and hold the Command and Shift keys down and drag either the video or audio portion of the selected clips to the left or right.
- 4 The selection of clips you're dragging will be more easily inserted in the middle of adjacent clips as you drag with snapping turned off, and the cut portion of each clip will be moved into the gap that's left behind by the clip(s) you're dragging. Drop the clip into the desired location when you're finished.

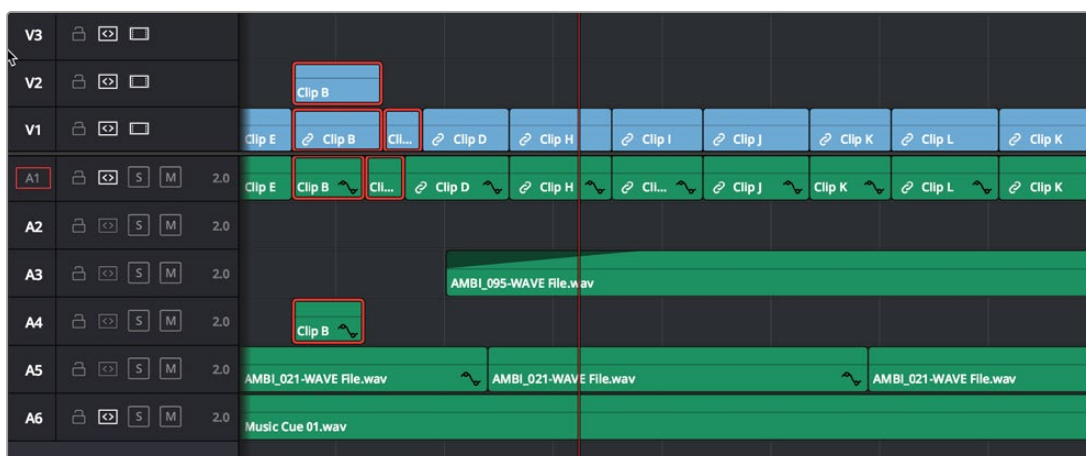


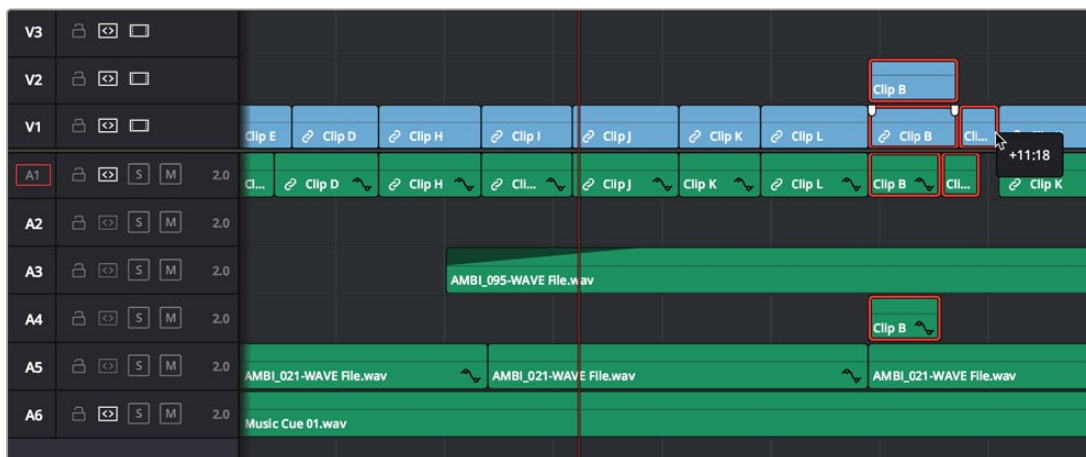
Before and after clip L being inserted into the middle of Clip K, which is split in half to make way for it

To shuffle insert multiple clips to another position in the Timeline:

- 1 Select all of the clips you want to move to another position on the Timeline.
- 2 Press and hold the Command and Shift keys, and drag the clips left or right. Make sure the item you click to drag is on the same track as the majority of clips you're rearranging; the item you click defines which track is used to guide the rearrangement of clips.

In the following example, the video item of Clip C is selected on track V1, so as it's dragged to the right, all clips on other tracks are rearranged according to the duration and location of clips B and C on track V1. As a result, clips on tracks other than V1 may be overwritten, or leave gaps, as necessary for the items on track V1 to be rearranged cleanly.





Before and after a group of clips being shuffled to the right. The clip that was clicked to drag defines how all other clips will be rearranged

To shuffle insert multiple clips from the Media Pool or Source Viewer into the Timeline:

- 1 Either select one or more clips in the Media Pool or open a clip in the Source Viewer.
- 2 Press and hold the Command and Shift keys, and drag the selection from the Media Pool or Source Viewer into the Timeline.
- 3 As you drag, the clips you're dragging will be inserted into the Timeline at the pointer location. Release the mouse to finish making the edit.

Splitting and Joining Clips

In many situations you may find yourself splitting clips (adding edits) in order to separate multiple clips that were inadvertently baked together, or to cut up clips into sections that you're planning on applying different effects to or grading differently.

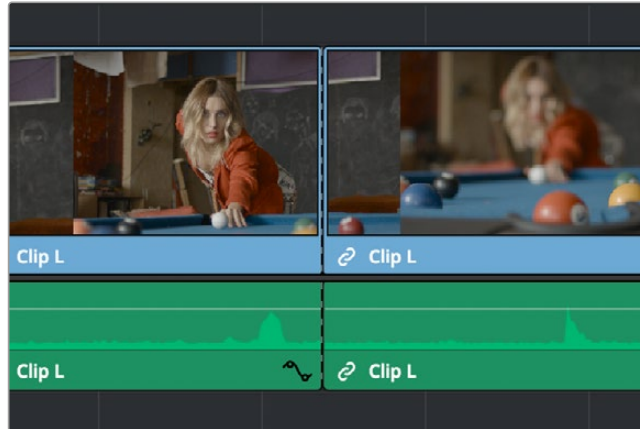
Methods of splitting and joining clips:

- **To split a clip once:** Drag the playhead to the frame where you want to split a clip, and press Command-Backslash (\) to split every clip on a track with Auto Select enabled.
- **To split many clips:** Click the Razor Edit mode button (or press B), and then click clips in the Timeline to split as many clips as you want.

TIP: Using the Blade Edit Mode or Split Clip command on a clip that's currently selected preserves the selection on the first half of the clip after cutting.

Through Edits

When you split a clip, a through edit appears to show that you currently have an edit with continuous timecode running from the outgoing to the incoming half. This is called a through edit, and is displayed with a dotted line running along its edge so you know that it's special.



A through edit seen in the Timeline

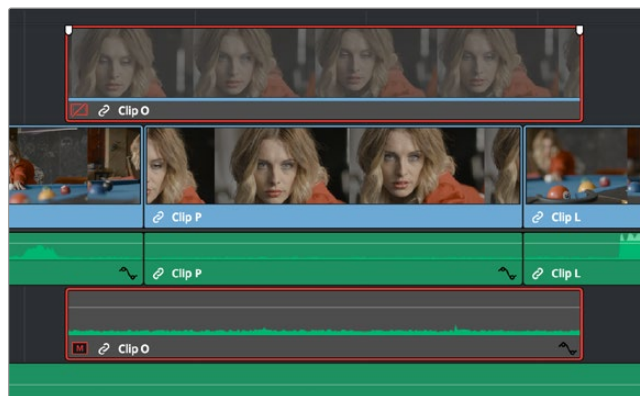
To eliminate a through edit, do one of the following:

- Select it in the Timeline, and press Delete.
- Right-click a through edit in the Timeline, and choose Delete Through Edit.

TIP: You can show an isolated list of every through edit in the Timeline by opening the Edit Index and choosing “Show Through Edits” in the Edit Index Option menu. Clicking any item in the list jumps the playhead to that through edit, making it easy to check all the through edits in a timeline to see if they’re necessary or not.

Enabling and Disabling Clips and Tracks

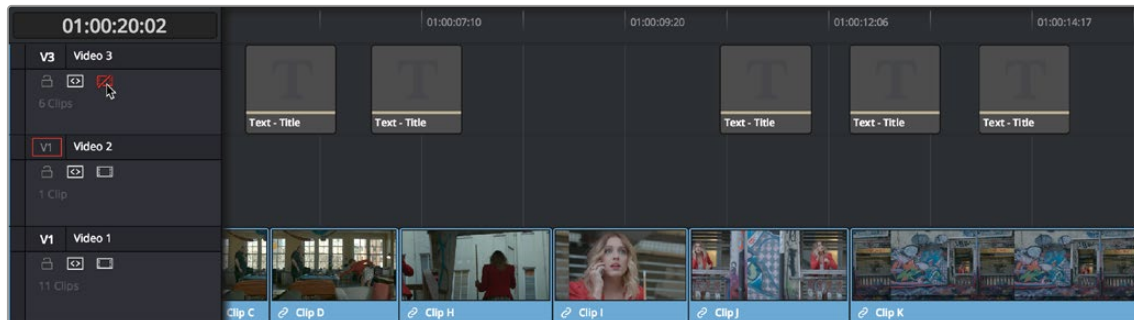
As you work in the Timeline, you’ll find there are times when you want to disable clips that you don’t want to appear during playback, without actually removing them from the edit. For example, you may decide to disable superimposed clips that are positioned as insert shots in the middle of a scene because of a client’s notes, but you don’t want to eliminate the clips because they might change their minds.



A disabled clip on track V2

In another example, you've edited a series of titles on track V3, so you need to disable track V3 in its entirety to output a textless version of the movie as a deliverable.

When a clip or track is disabled, clips within it appear dimmed, and these disabled clips don't appear in the Color page, and aren't output to tape or rendered to disk in the Deliver page until that track is re-enabled first.



Track V3 is disabled, making the Timeline textless as a result

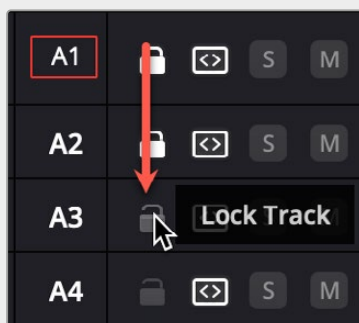
To disable/enable one or more clips in the Timeline:

- Select one or more clips, then right-click the selection and check or uncheck Enable Clip in the contextual menu, or press D to toggle a clip's enabled state.

To disable/enable an entire track:

- Click the track enable button.

You can change multiple Track Header controls in the Edit page (Lock Track, Visibility, Auto Track Selection, etc.) by simply clicking on an icon and dragging your pointer up or down across the track headers. You can do this for one type of control at a time and not across Audio and Video tracks.



Clicking on the Lock icon on A1, then dragging the pointer down to A3, will lock all audio tracks in between.

Copying and Pasting Clips in the Timeline

Clips can be cut, copied, and pasted in a variety of ways using standard keyboard shortcuts. You can cut or copy one clip or a selection of several, and you can also choose to copy or cut just the video or audio media of a clip. When pasting, you can paste to the same timeline, or to a different timeline if you want to move media from one to another.

Methods of doing simple cut, copy, and paste:

- **To cut one or more clips, leaving a gap:** Make a selection, and choose Edit > Cut (Command-X). The selected clip or clips are removed from the Timeline and stored in memory for pasting.
- **To ripple cut one or more clips and ripple the Timeline to close the gap:** Make a selection, and choose Edit > Ripple Cut (Command-Shift-X). The selected clip or clips are removed from the Timeline and stored in memory for pasting. All clips on tracks with Auto Select enabled will be rippled to the left to fill the gap left by the cut clips.
- **To copy one or more clips:** Make a selection, and choose Edit > Copy (Command-C). The selected clip or clips are left in the Timeline, but copies are stored in memory for pasting.
- **To paste one or more clips to the same track:** Move the playhead to the frame where you want the pasted selection to start, and then choose Edit > Paste (Command-V). By default, each copied clip is pasted onto the same track it was copied from.
 - Pasted clips will be added at the Playhead position.
 - Pasted clips overwrite any clips in that track that are in the way.
 - Pasted clips are automatically selected, ready for nudging left or right, or for other operations.
- **To paste one or more clips to a different track:** Pasting clips to a different track requires a slightly different procedure. Move the playhead to the frame where you want the pasted selection to start, then either Option-click any empty area on the track you want to paste the clip(s) to or Option-click the Auto Select control of that track to solo that track, and then choose Edit > Paste (Command-V).
 - Pasted clips will be added at the Playhead position.
 - Pasted clips overwrite any clips in that track that are in the way.
 - Pasted clips are automatically selected, ready for nudging left or right, or for other operations.

Paste Insert

Another paste command, Edit > Paste Insert (Command-Shift-V), lets you paste clips that you cut or copied via an insert edit, so that an edit is added at the position of the playhead to clips that are already in the Timeline, and all media to the right of the playhead is rippled farther right to make room for the clip or clips being pasted. As with all other ripple edits, only clips on tracks with their Auto Select control turned on are affected. Pasted clips are automatically selected, ready for nudging left or right, or for other operations.

Cut/Copy/Paste of Partial Clip Segments Using In and Out Points

By default, pasted clips will be added at the playhead position. You can use the Timeline's In and Out points to cut and copy partial segments of longer clips in various ways. This is a valuable technique for doing in-depth audio and dialog editing, although it's useful for copying partial segments of any kind of clip in the Timeline.

IMPORTANT: In order to use In and Out points for pasting, you must first go to Preferences > User > Editing, and check the box “Prioritize pasting to the in and out range.”

To cut or copy part of a clip to paste elsewhere:

- 1 Set In and Out points to isolate the part of the clip you want to cut or copy. You can use the Auto Select controls to include or omit clip segments on specific tracks while you do this.
- 2 Press Command-X to cut or Command-C to copy that clip segment.
- 3 Clear the In and Out points by pressing Option-X. Otherwise, you’ll paste the clip segment right back into the same place it started.
- 4 Move the playhead to the frame of the Timeline you want the pasted clip to start, and use the Paste or Paste Insert commands to paste the clip segment there. Pasted clips are automatically selected, ready for nudging left or right, or for other operations.

You can also use In and Out points to paste only a partial segment of a much longer clip that you’ve cut or copied.

To paste only part of a clip:

- 1 Select a clip, and press Command-X to cut or Command-C to copy that clip.
- 2 Set In and Out points to identify the region of the Timeline you want to paste into.
- 3 Use the Paste or Paste Insert commands to paste only as much of the Cut or Copied clip as will fit between the In and Out points you’ve placed. Pasted clips are automatically selected, ready for nudging left or right, or for other operations.

Copying and Pasting Clips to a Different Track

If all Auto Select controls on all tracks are turned on, clips are always pasted back to the same track they were copied from, starting at the position of the playhead. This is valuable for the many instances where you’ll find yourself copying and pasting clips you want to repeat, especially when doing audio editing.

However, if you want to paste the clips you cut or copied to a different track entirely, you need to use the Auto Select controls to specify which track you want to paste to. Here are the rules:

- You can force paste what you copied to a specific track by Option-clicking that track’s Auto Select control to solo it before pasting.
- When one or more Auto Select controls are disabled, then clips are pasted to the lowest-numbered track with an enabled Auto Select control.
- If you’ve copied clips on multiple tracks, clips on the lowest copied track will be pasted to the lowest Auto Select enabled track, and all other clips will be pasted to higher tracks, with new tracks automatically created, if necessary.
- If Auto Select is disabled on every single track, then a new track will be created above all other video tracks and/or below all other audio tracks, and the clip will be pasted into this new track, which has Auto Select turned on.

Audio Channels When Copying and Pasting Audio Clips

Copying and pasting audio has one other consideration. If you’re force pasting a clip into a different track, the track you solo the Auto Select control of could possibly be set to an audio channel mapping that doesn’t match the clips you’re pasting there. An example of when this would happen is if you copy stereo audio clips from a stereo track and paste them to a mono audio track.

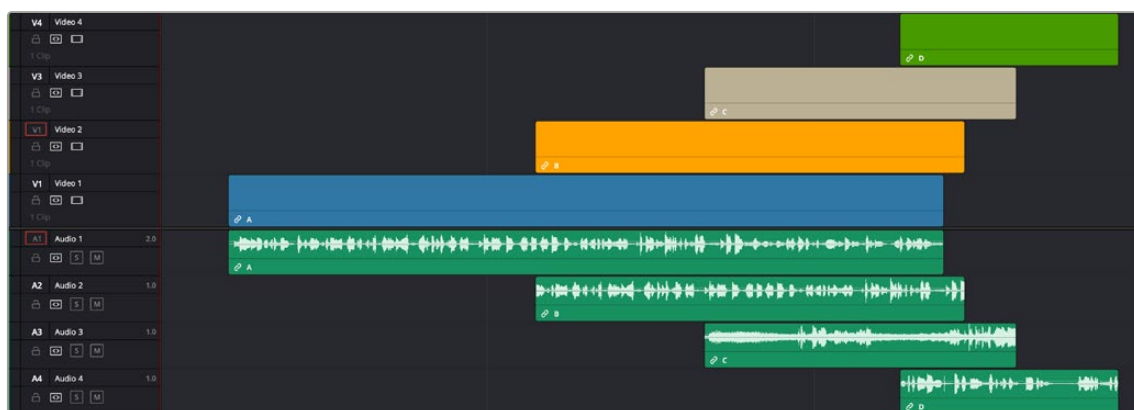
DaVinci Resolve allows you to do this, so you have the freedom of pasting audio clips to any track you want to. However, extra audio channels within clips that exceed the number of channels supported by the audio track they're on will be muted. Fortunately, this situation is easy to rectify. Simply right-click the track header of the problem audio track, and use the Change Track Type To submenu to change its channel mapping to one more appropriate to the clips you've pasted into it.

Auto Align Clips

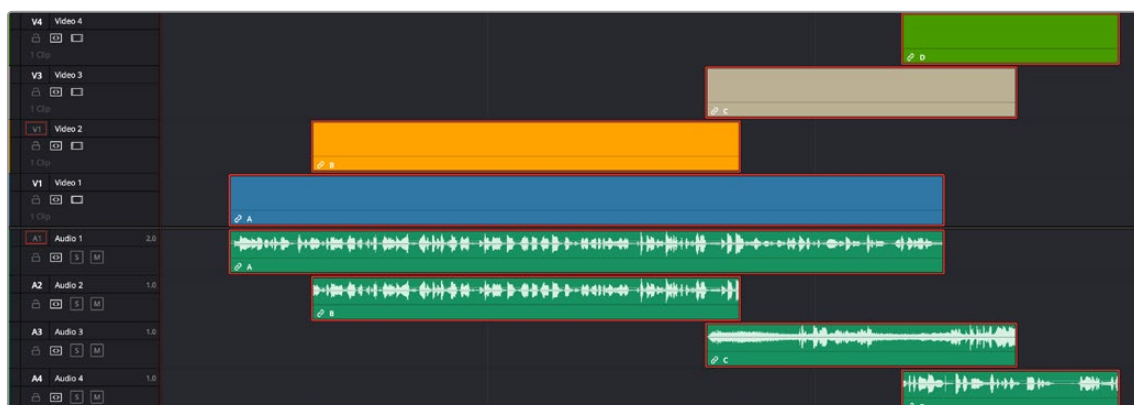
Auto Align Clips slides one or more selected clips to align with the timecode or audio waveforms of another clip that has matching timecode or audio. This function works for video clips, which can be aligned using timecode, and for audio clips, which can be aligned using either timecode or waveform matching. Waveform matching can also be used if you're working with audio/video clips.

You can only select one clip per video or audio track to align, and they all will align to whichever clip is on the lowest-numbered video or audio track. Clips selected that have no overlapping timecode or audio waveform will not be moved and left in their original position on the Timeline.

For example, clip A on track V1 overlaps with clip B on track V2, but not with clip C on track V3, or clip D on track V4. Selecting all clips and using Auto Align Edits will slide clip B to align with clip A, but clips C and D won't be moved because they don't overlap with clip A.

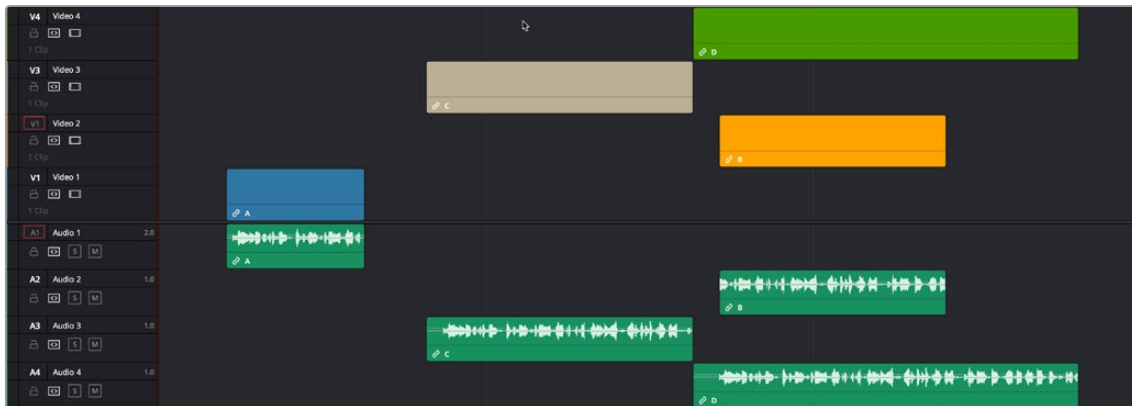


The original Timeline with clip A (blue) and clip B (orange) that have some overlapping audio but are out of sync

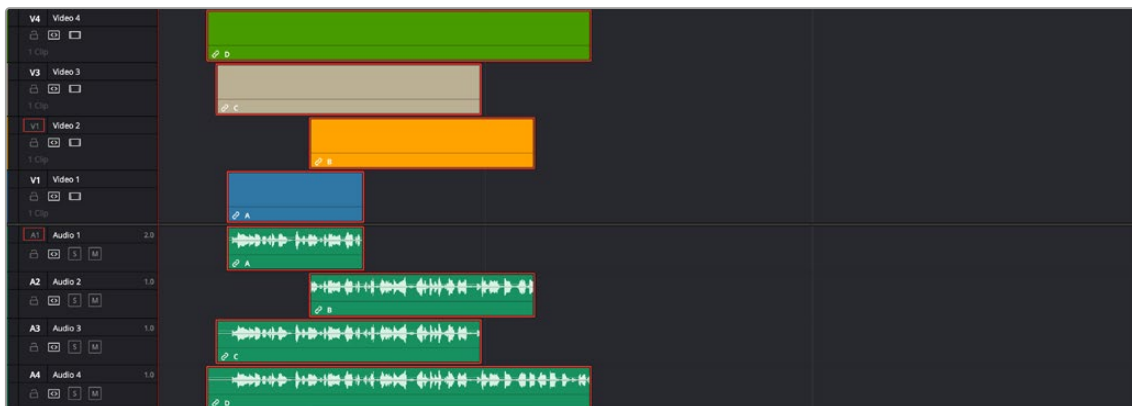


The same Timeline after the Auto Align command; clip B (orange) has been slipped to sync via audio waveform with clip A (blue). Clips C (tan) and D (green) were also selected, but because they had no overlapping timecode or audio, they remained in place.

In another example, clip A on track V1 overlaps with clips B, C, and D on tracks V2, V3, and V4 respectively. Selecting all clips and using Auto Align Edits will slide clips B, C, and D to be aligned with clip A.



The original Timeline with clip A (blue), and clip B (orange), C (tan), and D (green) all have overlapping audio, but are out of sync.



The same timeline after the Auto Align command. All clips are now in sync and the audio waveforms now match. All the other clips moved to clip A's (blue) position because it was the lowest clip on the Timeline.

To use Auto Align clips:

- 1 Arrange the clips you want to align with one another on the Timeline so that there's one clip per track. All clips will sync to the clip on the lowest track number. All clips must have overlapping timecode or audio waveforms.
- 2 Select every clip that you want to align (only one per track).
- 3 Right-click one of the selected clips, and choose an option from the Auto Align Clips submenu, either "Based on Timecode," or "Based on Waveform."

If Timecode is selected, the operation will be instant. If Waveform is used, a progress bar will appear showing you how long it will take DaVinci Resolve to analyze the selected audio waveforms before your clips are aligned. If you've selected clips that can't be aligned, a warning box will appear and tell you which clips had errors.

For video, this can be useful for multi-camera editing situations where you want to align an insert with the action of an alternate angle. For audio, this is useful for situations where you have multiple recordings of the same audio that you want to align for further editing.

NOTE: Waveform matching won't work for re-recorded audio, such as dialog that's been re-recorded using the ADR tools of the Fairlight page, as the correspondence between two waveforms must be precise for a match to be found.

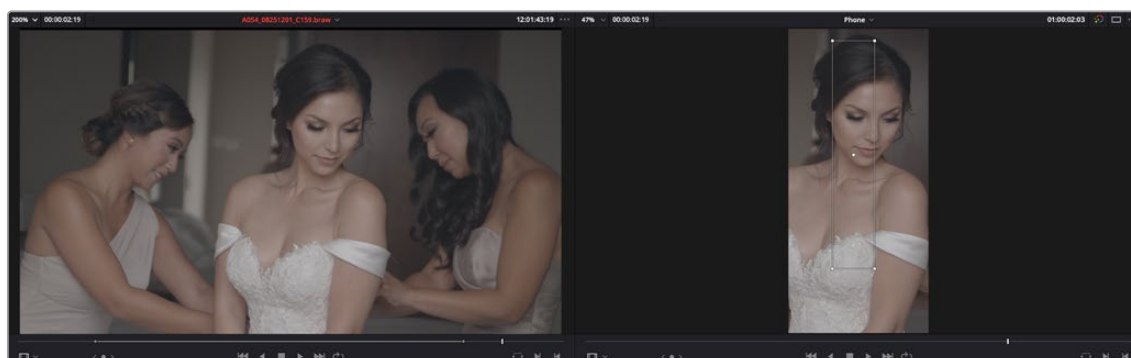
Duplicating Clips and Transitions in the Timeline

One or more clips can be duplicated by making a selection, and then Option-dragging the selected clips to another position and/or track in the Timeline. When duplicating clips in this way, you must hold the Option key down until you release the mouse button.

Individual selected transitions can also be duplicated by Option-dragging them to another edit point.

Smart Reframe (Studio Version Only)

The Smart Reframe feature in DaVinci Resolve makes it easier to quickly reframe material across extreme aspect ratio changes. It's useful for situations where you've shot a 16:9 horizontal video and find yourself needing to create a vertically-oriented 9:16 version for mobile phones and social media deliverables, or using 4:3 archival footage in a 2.39:1 widescreen movie. Smart Reframe can be used manually, or automatically executed using the DaVinci Resolve Neural Engine.



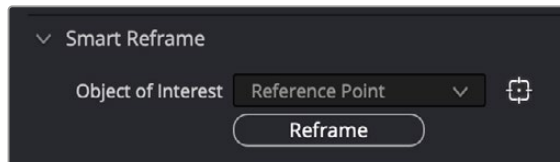
Smart Reframe in action, with the Reference Point bounding box active (right)

The Smart Reframe tool is found in the Sizing tab of the Inspector and is available in both the Cut and Edit pages.

To use the Smart Reframe tool:

- 1 Duplicate your timeline, right-click the Timeline and choose Timelines > Timeline Settings, and click Use Custom Settings to change the Timeline Resolution to the aspect ratio needed for delivery. Make sure that "Mismatched resolution files" is set to "Scale full frame with crop," and click OK.
- 2 Select one or more clips you want to reframe, and open the Inspector to the Sizing tab.
- 3 Open the Smart Reframe controls, leave the Object of Interest drop-down menu set to Auto (if you've selected more than one clip, Auto is the only setting available), and click "Reframe." DaVinci Resolve will analyze your footage and should automatically adjust each individual clip's position to a more aesthetically pleasing framing.

- 4 (Optional) If the “Auto” setting does not give you desirable results for a particular clip, you can manually select the main subject using the following steps.
- a) To manually select the subject area, choose “Reference Point” from the Object of Interest drop-down menu, and click the target icon just to the right of the menu. This automatically sets the Viewer mode to Smart Reframe, exposing the onscreen controls for choosing a reference.
 - b) Drag the Reference Point bounding box around the main subject of interest in the frame. You may use the Transform controls directly above in the Inspector to move the source clip around if your subject is outside the current framing.
 - c) Click “Reframe.”

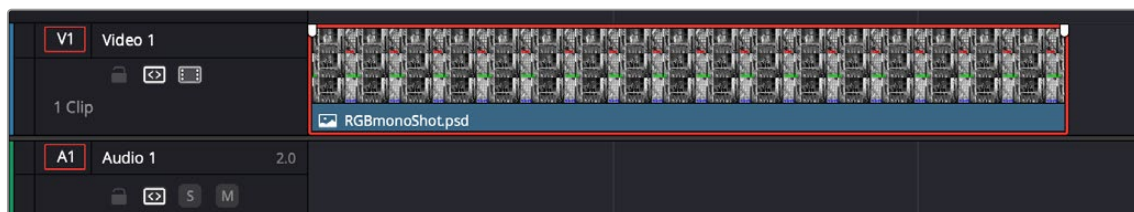


The Inspector's Smart Reframe controls showing the manual reference point selected

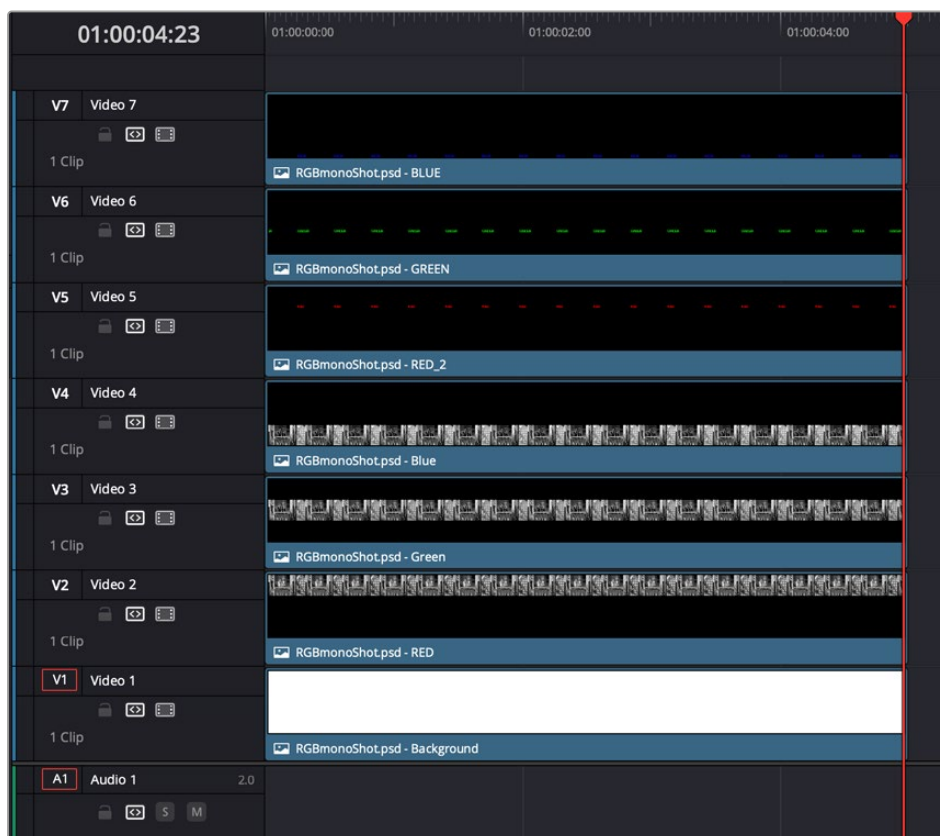
DaVinci Resolve locks onto and, if necessary, tracks your subject using the reference you've selected, automatically panning and scanning the original clip as needed to keep the reference within the new aspect ratio. While involving a bit of manual adjustment, this function still dramatically reduces the time involved in pan and scanning footage by manually adjusting and keyframing the sizing controls.

Split Photoshop PSD Layers into Individual Timeline Tracks

You can right-click on an Adobe Photoshop Document (.psd) in a timeline and select Split PSD Layers in Place to expand each layer of the file into its own track. The number of new tracks created will correspond with the number of layers in the file.



You can split Photoshop PSD layers into individual tracks by right-clicking on them and selecting Split PSD Layers in Place.



The results of splitting a PSD file.

DaVinci Resolve reads and splits rasterized flattened layers, so some steps need to be taken when preparing the underlying PSD asset before import. Merging layers in Photoshop to create flattened layers is generally a manual process. The intention is to simplify the image structure, bake Photoshop-specific effects and rasterize fonts to ensure compatibility with DaVinci Resolve. Here is a breakdown of the process:

Rasterize transparency and effects

If your layers/fonts contain effects like shadows or gradients, merging them can sometimes alter their appearance. Consider rasterizing before merging. To do this, select one or more layers, right-click and choose Rasterize Layer Style. This is recommended for text layers as well.

Merge and flatten similar layers

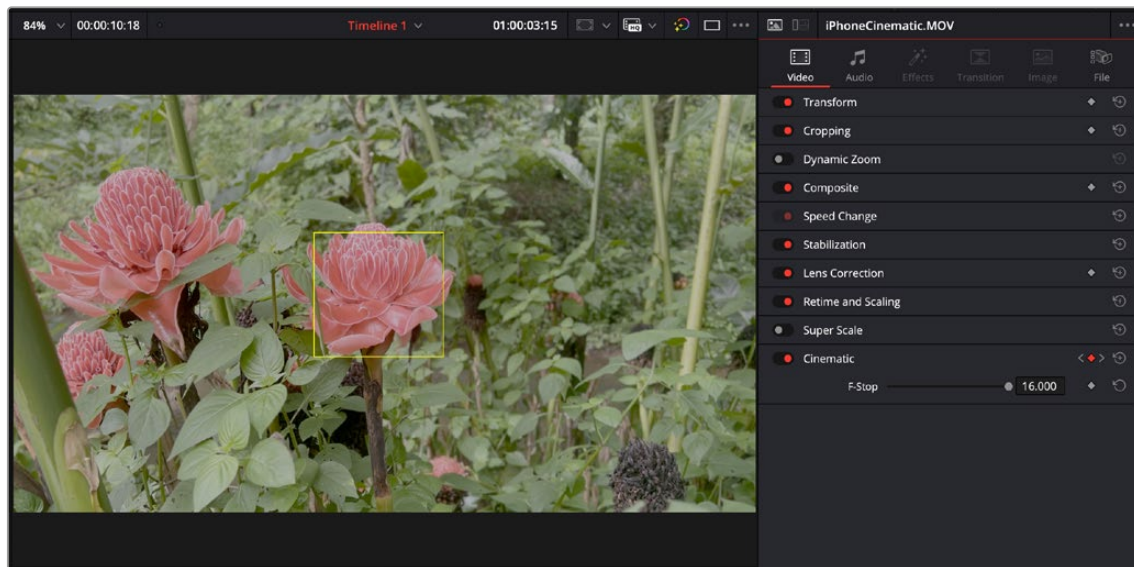
Arrange similar layers together and select the layers you want to merge. Right-click on the selected layers and choose Merge Layers (Cmd/Control+E) to combine the selected layers into one 'flat' layer. Depending on the complexity of the Photoshop document, consider using the option: File > Scripts > Flatten All Layer Effects.

Name layers

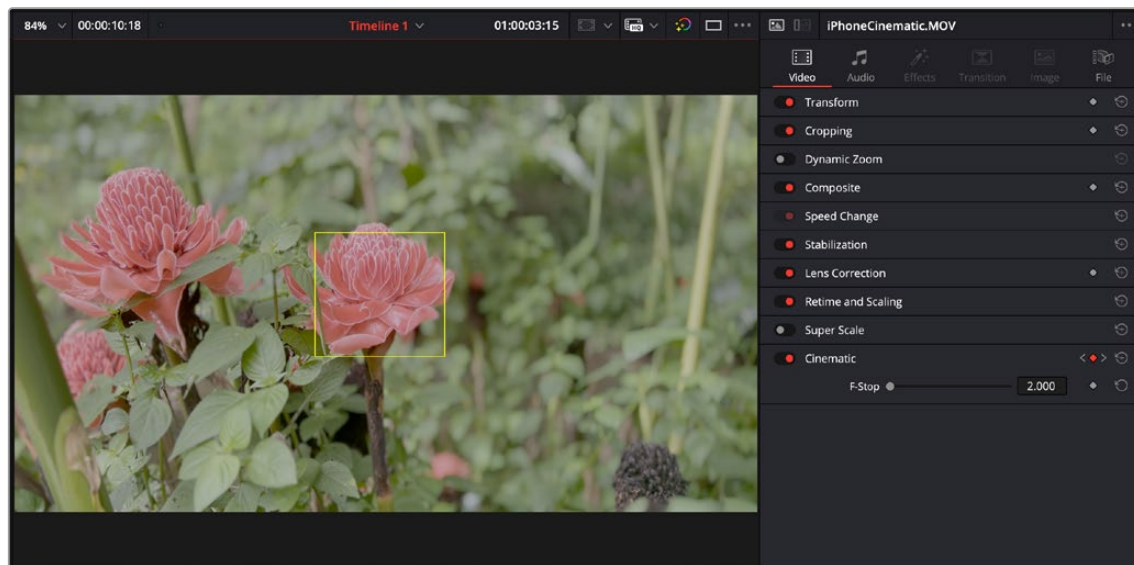
Rename the layers to describe the underlying image, effect or element. Naming the flattened layers at this stage allows DaVinci Resolve to use the names in the split clips.

Apple Cinematic Mode (Mac OS Only)

Recent models of Apple iPhone have a Cinematic mode that use lidar to emulate the depth of field characteristics of a lens iris. You can use this mode to artificially create shallow depth of field looks or to pull focus from one focal point to another. This functionality can be accessed directly in the Edit Timeline of DaVinci Resolve.



Apple Cinematic mode with an F-Stop of 16; note the in focus background. The Yellow square denotes the focus point.



Apple Cinematic mode with an F-Stop of 2; note the blurred background. The Yellow square denotes the focus point.

Using Apple Cinematic Mode in DaVinci Resolve

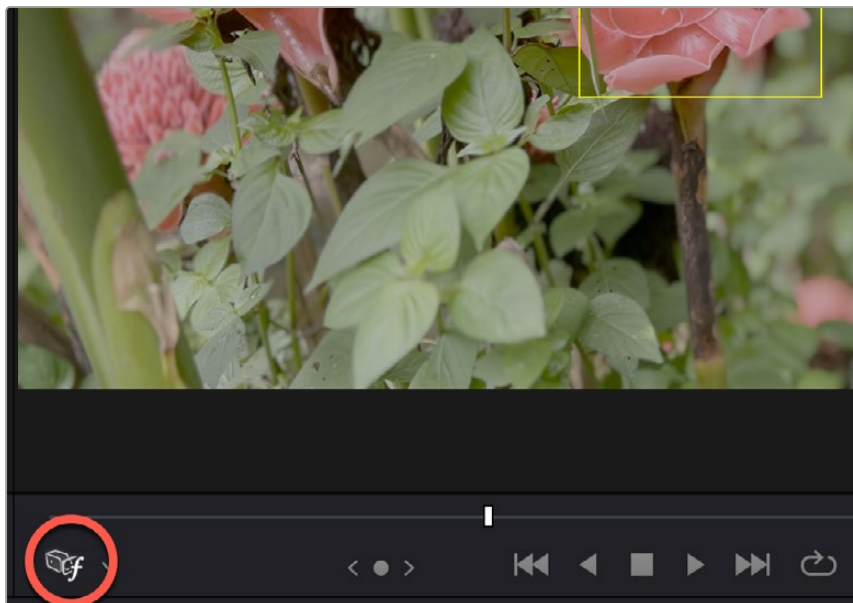
Using Cinematic footage recorded on an iPhone takes some additional steps to bring into DaVinci Resolve and is only available on the MacOS version.

To import iPhone Cinematic footage into DaVinci Resolve:

- 1 On your iPhone, the camera mode must be turned to Cinematic before you record the video.
- 2 Once the video is finished, find the video in the Photos app and press the Share icon.

IMPORTANT: In the Share interface, select “Export Unmodified Original.” Sharing any other way (like AirDrop) will remove the cinematic information from your video.

- 3 Choose a location to save to on your phone in the Files interface.
- 4 From the Files interface, find the cinematic video, tap it, and then select the Share button again.
- 5 Now export the cinematic video via AirDrop or other method to your computer.
- 6 Open DaVinci Resolve and import the cinematic video into the Media Pool as normal.



Put the Viewer Overlay in Cinematic mode (circled) in the drop-down menu.

To use the Cinematic tools in DaVinci Resolve:

- 1 Put a Cinematic Video clip on the timeline and select it.
- 2 In the Viewer Overlay, select Cinematic from the drop-down menu.
- 3 In the Video Inspector, the Cinematic toolset should appear at the bottom of the list.
- 4 Choose a recorded focus point in the Viewer, designated by a yellow rectangle.
- 5 Use the F-Stop controls to adjust the depth of field.

Scene Cut Detection on the Timeline

If you need to break down a previously edited video into its component clips for re-editing or color correction, you can do so directly in the Timeline. Using the DaVinci Neural Engine, DaVinci Resolve can automatically analyze and split up an edited video into individual clips.

Timeline Scene Cut Detection is also available in the Cut page. If you prefer, you can continue to use the original Scene Cut Detection tool found in the Media Pool.

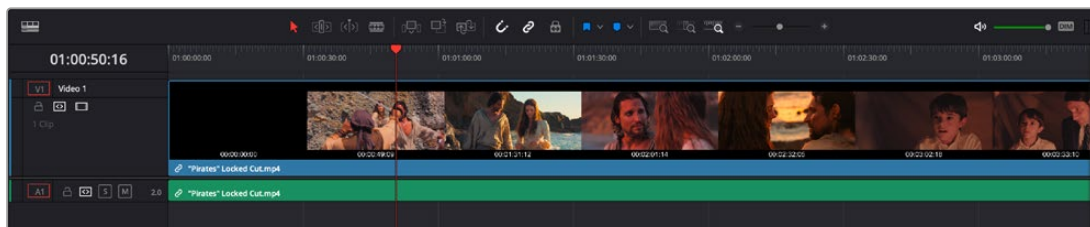
To use Scene Cut Detection on the Timeline:

- 1 Select one or more clips you want to split on the Timeline. Alternately, you can limit Scene Cut Detection to just a portion of a clip by setting In and Out points on the Timeline around the section you want to analyze.
- 2 Choose Timeline > Detect Scene Cuts.
A dialogue box appears, “Detecting scene cuts in clips x of x.” This process can take some time, depending on the length, number, and complexity of the clips you’ve selected. When the Scene Cut Detection has finished, the clip you selected will be broken up into a number of through edits that now can be used as independent clips.

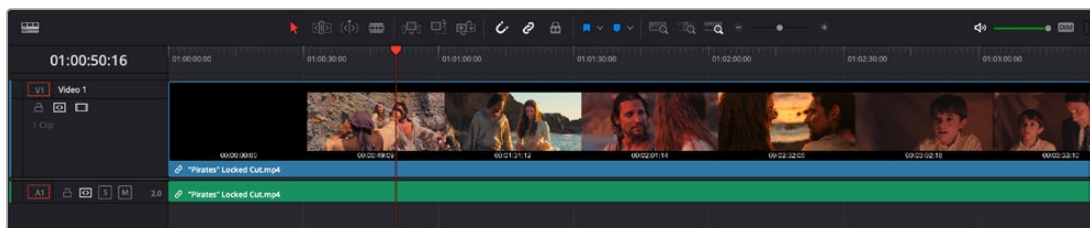
Checking and Fixing Your Results

If the Neural Engine has made an error, you can fix it manually by navigating to the cut using the Up and Down Arrow keys to go back and forth in the Timeline, and by then doing one of the following:

- **To remove a Cut:** Click the through edit to select it, and press the “Delete” key.
- **To make a New Cut:** Place the timeline indicator at the cut point, and choose Timeline > Split Clips (Command-).)



A single clip of a finished edit, consisting of multiple cuts before the Detect Scene Cuts command

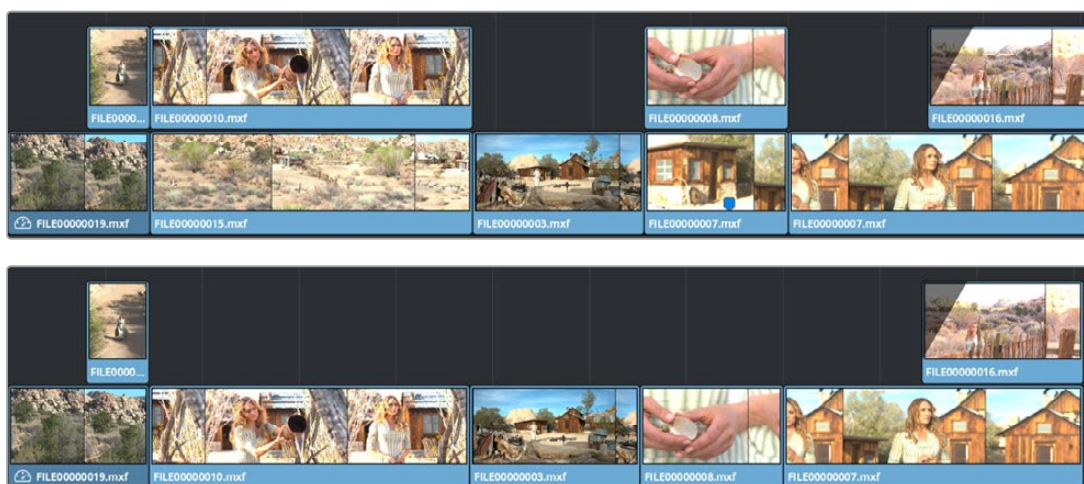


Multiple individual clips extracted from the edited clip via Detect Scene Cut; the operation has been contained by the In and Out points, and one of the resulting through edits has been highlighted in green.

Clean Up Video Tracks

While it's convenient to be able to edit overlapping clips on multiple video tracks to try slipping clips back and forth, or to stack multiple takes or versions of a VFX clip, there comes a time when all those unnecessary clips take a toll on the visual organization of your timeline, not to mention your colorist's sanity. For this reason, a trio of commands for cleaning up your timeline have been added to the Timeline > Clean Up Video Tracks menu. These are:

- **Flatten Unused Clips:** All superimposed clips with In and Out points that are aligned with clips below them are moved down to track V1, so long as they don't have any kind of opacity, composite mode, transition, or fade effect applied to them making them a compositing effect.
- **Disable Unused Clips:** All clips that are underneath superimposed clips that don't have any kind of opacity, composite mode, transition, or fade effect applied to them are disabled.
- **Change Unused Clips Color:** All clips that are underneath superimposed clips that don't have any kind of opacity, composite mode, transition, or fade effect applied to them have their color changed to whatever you select.



(Top) The original timeline, (Bottom) The Flatten Unused Clips command is used to move superimposed clips with In and Out points that match other clips underneath them to track V1 to simplify the Timeline for future work